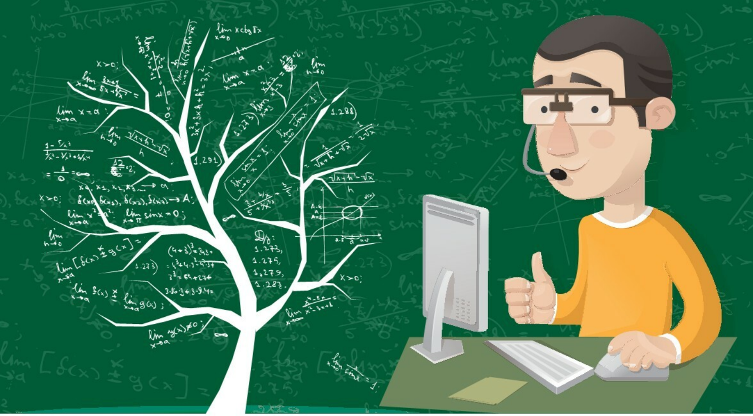
Get Introduced to Algebra with Maxima

Maxima is a computer algebra system derived from the earlier Macsyma which in turn, was written in Lisp. This article, the 14th in the series on the mathematical journey through open source, introduces us to algebra, using Maxima.



The algebraic capability of Maxima is what sets it apart. We have worked through basic shell math, bench calculator *bc*, and finally, the ultra capable Octave. But none of these systems could do algebra with symbols, also referred to as symbolic or analytic mathematics. Let's walk through a few examples to demonstrate what this means.

Getting started

The Maxima command line can be started in the shell by typing *maxima*. The *-q* option suppresses the initial welcome message. Typing *quit();* on the maxima command line quits Maxima. Typical Maxima commands are terminated either with a semicolon (;) or a dollar ($) - the latter suppressing any print from the command.

Outputs are numbered and prefaced by *%o*. Inputs are also numbered, and prefaced by *%i*. Here are a few factorisation examples:

```
$ maxima -q
(%i1) factor(x^3 - 1);

(%o1)                      (x^2 - 1) (x  + x + 1)
(%i2) factor((x + 1)^4);

(%o2)                              (x^4 + 1)
(%i3) expand(%o2);

(%o3)                   x^4  + 4x^3  + 6x^2  + 4x + 1
(%i4) factor(6!);
```